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An innovation resistance theory perspective on metaverse NFTs: The moderating effect of metaverse usage motivation

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ABSTRACT

This study uses innovation resistance theory (IRT) to investigate why metaverse users are hesitant to purchase metaverse NFTs. We also broaden the theory further by exploring the moderating role of metaverse usage motivation (communication, experience of new worlds, self-expression, and economic activity). As part of the study, data were collected from 500 metaverse users through an online survey. Subsequently, to verify the hypotheses of the study, structural equation modeling (SEM) was carried out using AMOS software, and the moderating effect was verified using SPSS Process Macro Model 1. The results of the study suggest that usage, value, risk, tradition, and image barriers had a significant negative effect on purchase intention of metaverse NFTs. In addition, the findings of this study discuss various moderating effects of metaverse usage motivation on the relationship between them. So, these results are expected to provide a theoretical framework for future IRT research. They also provide effective marketing communication strategies for metaverse platform practitioners and marketers by verifying users' innovation resistance to NFTs, which may become the basis for a major economic system in the metaverse in the future.

1. Introduction

Human lifestyle and quality of life in the 21st century have made major shifts owing to three major technological advances: PCs, the Internet, and smartphones. In addition, since the coronavirus pandemic, metaverses have become the next major paradigm shift in the world's attention: In 2023, the global metaverse market size was estimated to be $94.1 billion U.S. dollars, and it was projected to reach $130.5 billion in 2024 and over one trillion U.S. dollars by 2030 (Statista, 2024).

Along with the metaverse's remarkable growth, nonfungible tokens (NFTs) are also gaining increased attention. NFTs are a category of digital assets built on blockchain technology, characterized by their distinct and unique properties (Raman & Raj, 2021). These tokens represent various entities, such as graphics, artwork, digital clothing, vehicles, virtual land, user avatars, and virtually any other physical entity that can have its presence in a digital realm (Basu et al., 2025). In the metaverse, NFTs have become a key element, transforming how people understand ownership and value in the digital world (Belk et al., 2022). NFTs are being used in various fields, and they are highly likely to become a key element of the metaverse's economic system in the future, based on the verification of digital asset ownership. The metaverse is anticipated to include a 'virtual economy' that integrates NFTs and digital currencies (Jafar et al., 2023; Wiles, 2022).

Although both the metaverse in general and NFTs more specifically experienced an accelerated development during the coronavirus pandemic lockdowns, there are arguments that the metaverse is still premature and unreal. Researchers from the Pew Research Center and Elon University surveyed a group of 624 experts on their opinion on metaverse. In response to the question, "Do you expect the metaverse to become fully immersive and part of daily life for more than 500 million people worldwide by 2040?" 54 % of the experts said 'yes', while 46 % said 'no' (Pew Research Center, 2022). So, opinions and claims about the metaverse are extremely polarized, and the current corporate and public opinion that the metaverse is popular and that people need to join the trend or be left behind will only lead to resistance from the public.

Similarly, NFTs also remain a debatable subject of whether they are a financial innovation or a fantasy. The NFTs and metaverse market experienced a significant decline recently, with decreasing transaction volumes and market capitalization, which led many projects to struggle to maintain momentum and investor interest, contributing to the overall contraction of the market (Jacobides et al., 2024; Kumar et al., 2025). Additionally, the legal and institutional framework for possessing and trading NFTs is still underdeveloped, raising concerns about taxation, money laundering, and other issues. The rise and fall of the NFT market

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can be explained by the phenomenon of the Gartner technology hype cycle, where new technologies or trends initially generate a surge of excitement and interest, leading to rapid adoption and price increases (Chun, 2023; Perez and Kreinovich, 2018). However, once the hype declines, the market is expected to experience a congruent decline in interest and prices, ultimately stabilizing and reaching a more mature level of adoption.

Despite the attention NFTs have gained, the majority of discussions about NFTs have focused primarily on their technical aspects (Mamidala and Kumari, 2023; Park et al., 2022), while overlooking their significant role in transforming brand marketing strategies to engage consumers and influence purchasing behaviors (Kumar et al., 2025). Moreover, metaverse retailing and purchasing behaviors have been underexplored (Jafar et al., 2023). To address this gap, this study applies innovation resistance theory (Ram & Sheth, 1989), which offers a framework for understanding user resistance to innovation, to identify barriers that negatively affect metaverse NFT purchase intention. Subsequently, by incorporating users' usage motivations into the research model, we examine how these motivational factors moderate the relationship between user resistance and purchase intention. This approach contributes to filling the existing gap in the literature, which has primarily focused on the technical aspects of the metaverse and metaverse NFTs, and extends the application of Innovation Resistance Theory. Furthermore, this study offers practical insights for metaverse marketing professionals and metaverse platform operators to develop goal-oriented strategies that overcome barriers and promote NFT purchase intentions in the metaverse environment.

2. Literature review

2.1. Metaverse NFTs

Recently, NFTs in the metaverse have been attracting attention as a way to expand the metaverse economic ecosystem and business model. As described earlier, NFTs may significantly contribute to the growth of the metaverse by functioning as reliable means to authenticate ownership and originality of digital assets held by consumers (Seong et al., 2021). Furthermore, NFTs can be resold by buyers through blockchain-based smart contracts, which facilitates secure transactions and enhances market liquidity (Chohan & Paschen, 2023; Sung et al., 2023). As the NFT-based metaverse ecosystem expands and enables NFT intercompatibility between different metaverses, it will become more valuable as a part of the economy beyond just games (Software Policy & Research Institute, 2021).

To understand NFTs in the metaverse market economy, it is necessary to understand the metaverse revenue structure and business model. The current form of metaverse profits can be categorized into B2B (business to business), B2C (business to consumer) and C2C (consumer to consumer). From a macro perspective, metaverse profitability models tend to be B2B models in which a platform provider creates and sells a metaverse that a company needs, or sells or rents virtual space within the metaverse. Importantly, such virtual spaces are often structured as NFTs, meaning that ownership and transaction records are securely stored on the blockchain. This allows virtual real estate in the metaverse to be bought, sold, or transferred in a trustworthy manner, thereby expanding the scope of NFT applications beyond digital items to include spatial assets as well. Metaverse profitability models from a micro perspective can be labeled as B2C and C2C, where platform providers sell content or NFTs to users or trade among users.

Seok (2021) presented the main cases and business models in an analysis of the metaverse ecosystem and classified four types of metaverse businesses: content creation and sales, media mediation commissions, marketing commissions, and subscription fee. First, related to content creation and sales, a platform provider sells content to users or takes a transaction fee for users' right to create and sell content. For example, in Roblox, users create and sell games, and Fortnite collaborates with famous brands to create and sell avatar clothing items in the form of NFTs. These profitability models take the form of B2C and C2C. Second, platform providers take media mediation commissions by embedding other media platforms within the metaverse platform. For example, Fortnite sold tickets to the premiere of a new song by the popular idol group BTS, and by selling the ability for avatars to follow the choreography generated revenue for both the platform provider and BTS's management company, Big Hit Entertainment. These profitability models take the form of B2C and B2B. Third, marketing commissions are earned when advertisers market on social media platforms to generate revenue. Under a B2B model, businesses can attract customers into the virtual world spaces of the metaverse platforms to enter stores and earn store fees. Fourth, subscription fees represent a B2C model that relies on consumers to subscribe to services such as Netflix, Amazon prime video, and Apple TV+.

As we discussed above, both NFTs and metaverse platforms have been the subject of ongoing debate about whether they are an innovation or a fantasy. The legal and institutional framework for possessing and trading NFTs is still incomplete, and issues such as taxation and money laundering have also arisen. Despite the NFT market experiencing significant and unpredictable fluctuations, there is a lack of clear evidence explaining its downturn, and scholarly research along with statistical data on user interaction within NFT-based metaverse environments remains limited and underdeveloped (Albayati et al., 2023). A major barrier to the development and growth of NFTs within metaverse platforms lies in the transformation of user perception. In NFTs, the data source is literally a digital asset that does not exist in the real world and does not equate to a real-world source: For users who are accustomed to the existing real economy to accept the innovation of digital assets, it is necessary to change users' perceptions and acceptance of the value of digital assets. Therefore, this study focuses on metaverse users' perceptions and attitudes toward metaverse NFTs and aims to contribute to the stream of research on perceptions and attitudes toward metaverse NFTs, as summarized in (Table 1).

The major prior studies on metaverse users' perceptions and attitudes toward NFTs have primarily focused on positive usage motivations and brand effects, with limited systematic exploration of the factors that hinder NFT purchase intentions. In response, this study seeks to address this theoretical gap by focusing on the overlooked aspect of user resistance. Furthermore, by identifying not only the factors that promote acceptance but also those that act as barriers to the adoption of metaverse NFTs, this study aims to provide differentiated and practical strategic implications.

2.2. Innovation resistance theory

Ram and Sheth (1989) proposed innovation resistance theory as a framework for studying consumer resistance or barriers to innovation adoption. Innovations that deviate from current beliefs and the present state are likely to lead to resistance (Ma & Lee, 2019; Seth et al., 2020). Consumer resistance is an important variable that determines the success or failure of new technological innovations (Kaur et al., 2020a; Talwar et al., 2020).

Many researchers have studied innovation resistance (Ram, 1987; Rogers, 1995; Szmigin & Foxall, 1998), starting with Sheth's 1981 psychological innovation resistance. Ram and Sheth (1989) identified three factors that influence consumer resistance to new products and developed an innovation resistance model to represent the relationships between them. These three factors were perceived innovation characteristics, consumer characteristics, and characteristics of the diffusion structure, and they are influenced by cultural, social, and situational factors. In response, companies must repeatedly modify their innovations to meet the needs of consumers (Fig. 1).

Sheth and Ram (1987), sought to understand why people resist innovations and identified five function-related barriers to innovation: specialization, operational, resource, regulatory, and market access.

Table 1
Summary of research on users' perceptions and attitudes toward metaverse NFTs.

Reference	Research aim	Factors	Findings and contributions
Albayati et al. (2023)	To examine and quantify the factors influencing user engagement in the NFT metaverse using an extended theory of planned behavior model	Attitudes toward behaviors, Subjective norms, Perceived behavioral control, Market impact, Trust impact, Social impact, Technical impact, Regulatory impact, Intention, Behavior	Significant positive effects of theory of planned behavior components (attitude, subjective norm, perceived behavioral control) on behavioral intention; Significant impacts of trust, social, technical, market, and regulatory factors on intention and behavior
Lee et al. (2023)	To examine how branded non-fungible token attributes influence brand attitude, brand commitment, purchase intention, and active engagement in the metaverse using the stimulus-organism-response framework	Scarcity, Financial value, Prestige, Uniqueness, Originality, Communication consistency, Brand attitude, Brand commitment, Purchase Intention, Active engagement	Significant positive effects of scarcity, financial value, prestige, uniqueness, originality, and communication consistency on brand attitude; brand attitude significantly predicted brand commitment, branded non-fungible token purchase intention, and active engagement
Sung et al. (2023)	To explore psychological mechanisms driving consumer behavior toward NFT luxury fashion items in the metaverse across cultures	Cultural dimension (Individualism, Collectivism), Metaverse appearance importance, Social value of NFTs, Economic value of NFTs, Luxury brand NFT authenticity, Luxury brand NFT scarcity, Anticipated gains of buying luxury brand NFTs, Anticipated losses of not buying luxury NFTs, Luxury Brand purchase intentions in metaverse, Ideal self-image congruence in metaverse, Luxury brand purchase intentions in reality	Significant effects of economic and social value on perceived authenticity and scarcity; Positive influence of authenticity and scarcity on purchase intention; moderating effects of cultural orientation and gender
Xie et al. (2023)	To investigate how perceptions of branded NFTs affect attitudes toward NFTs, brand loyalty, and brand purchase intention among Millennials and Generation Z	Perceived usefulness, perceived ease of use, perceived playfulness, perceived risk, perceived scarcity, NFT attitude, NFT norms, NFT buy, Brand attitude, Brand loyalty, Purchase intention	Perceived risk negatively affects attitude toward non-fungible tokens; Attitude and norms influence willingness to purchase; Willingness to purchase and brand attitude enhance brand loyalty and purchase intention
Lee & Shen (2024)	To investigate how different forms of gratification influence NFT creators' identification and their subsequent creation and consumption behaviors on NFT platforms	Artistic value, Content richness, Extrinsic rewards, Intrinsic rewards, Self-image expression, Recognition, NFT identification, Community identification, Creation, Consumption	Significant positive effects of artistic value, content richness, extrinsic and intrinsic rewards, self-image expression, and recognition on NFT creation and community identification, which in turn significantly impact creation and consumption behaviors
Sangsawang (2024)	To identify determinants of NFT purchase intention by integrating TPB and TAM models	Perceived usefulness, Perceived ease of use, attitude, Subjective norms, Perceived behavioral control, Perceived risk, Purchase intention	Significant positive effects of perceived usefulness, perceived ease of use, attitude, subjective norms, and perceived behavioral control on purchase intention; Significant negative effect of perceived risk on purchase intention

They also identified five consumer-related barriers to innovation: usage, value, risk, tradition, and image. Notably, the comprehensiveness of Ram and Sheth's (1989) innovation resistance theory makes it a useful framework for investigating user resistance to innovation (Ma & Lee, 2019). Existing theoretical frameworks, such as innovation diffusion theory and technology adoption models, do not focus on user resistance to innovation (Gupta & Arora, 2017).

According to innovation resistance theory, consumer resistance is classified as active or passive (Heidenreich & Handrich, 2015): Active resistance emerges from a direct connection to the innovative feature, whereas passive resistance emerges from a conflict between the innovation and the user's existing beliefs (Yu & Chantatub, 2015). Furthermore, active resistance is an undesirable evaluation of an innovation (Nabih & Poiesz, 1997), a perception that it failed to meet expectations (Laukkanen, Sinkkonen, & Laukkanen, 2008); passive resistance occurs as a refusal to evaluate an innovation (Bagozzi & Lee, 1999). Lastly, active resistance is explored through functional barriers (usage, value, and risk) and passive resistance is explored through psychological barriers (tradition and image) (Yu & Chantatub, 2015).

Researchers have widely applied innovation resistance theory to technology-based topics (Table 2), such as online shopping (Lian et al., 2012; Lian & Yen, 2014), mobile banking (Yu & Chantatub, 2015), augmented reality mobile games (Oktavianus et al., 2017), mobile shopping (Gupta & Arora, 2017), internet banking (Borraz-Mora et al., 2017), mobile social commerce (Hew et al., 2019), online travel agencies (Talwar et al., 2020), mobile payment solutions (Kaur et al., 2020a), food delivery applications (Kaur et al., 2020b; Verma et al., 2023), mobile ticketing applications (Chen et al., 2022), mobile payment (Migliore et al. (2022), service robots (Lee & Kim, 2022), fitness apps (Chakraborty et al. (2023), WhatsApp payment system (Uddin et al., 2025), and metaverse products (Nie et al., 2025). The widespread application of innovation resistance theory across these emerging technologies suggests that this theoretical framework can also be effectively utilized to understand user resistance toward metaverse NFTs.

Researchers have established that certain functional and psychological barriers negatively affect users' intentions and behaviors. According to Huang et al. (2021), who conducted a qualitative study on innovation resistance, most researchers have not provided detailed strategies for how to overcome this resistance. Many companies face a

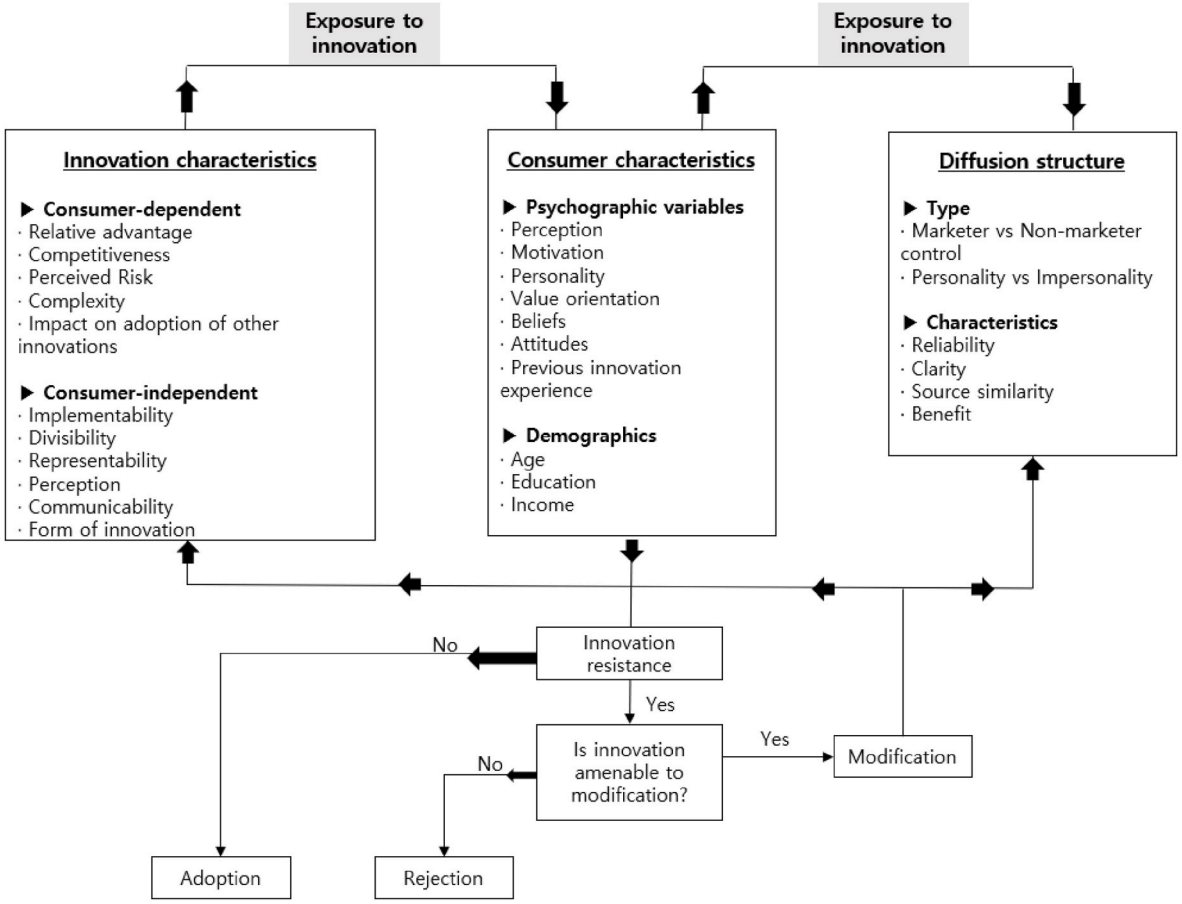


Fig. 1. Modified Innovation Resistance Model (Based on Ram and Sheth, 1989).

high probability of failure when introducing new products and services to the market due to consumer resistance to innovation (Ram, 1987; Ram & Sheth, 1989). Therefore, it is necessary to find out what factors lead to innovation resistance from the user side for metaverse NFTs, which are expected to bring about a paradigm shift in future lifestyles with global attention.

With this study, we explored consumer resistance to metaverse NFTs utilizing functional (usage, value, risk) and psychological (tradition, image) barriers as key variables proposed by Ram and Sheth (1989) and utilized in many previous studies since then. We then suggest marketing communication strategies for overcoming these barriers. Based on our literature review, we propose the following hypotheses regarding the impacts of metaverse users' functional (usage, value, and risk) and psychological (tradition and image) barriers to NFT usage on their intention to purchase the new digital asset called metaverse NFTs.

Research Hypothesis 1. Users' usage barriers to using metaverse NFTs will negatively affect their metaverse NFT purchase intention.

Research Hypothesis 2. Users' value barriers to using metaverse NFTs will negatively affect their metaverse NFT purchase intention.

Research Hypothesis 3. Users' risk barriers to using metaverse NFTs will negatively affect their metaverse NFT purchase intention.

Research Hypothesis 4. Users' tradition barriers to using metaverse NFTs will negatively affect their metaverse NFT purchase intention.

Research Hypothesis 5. Users' image barriers to using metaverse NFTs will negatively affect their metaverse NFT purchase intention.

2.3. Uses and gratification theory and metaverse usage motivation

Research on uses and gratification (U&G) theory can be divided into analyses of general versus specific media usage motivations (Choi, 2007), and the theory has provided an understanding of and theoretical implications for new media environments (Han et al., 2011). U&G theory has suggested a new approach to new media and a useful conceptual framework for understanding the changing media environment (Kaye & Johnson, 2002; Lin, 1999; Morris & Ogan, 1996; Papacharissi & Rubin, 2000). Since the expectation of being satisfied with media is related to evaluations of the medium's characteristics, usage motivation factors likely vary depending on the medium's properties (Kim & Park, 2007). U&G theory plays an important role in understanding the characteristics of new media and users' usage behaviors, attitudes, and motivation and is an effective theoretical framework for studying metaverse usage behaviors.

With the global interest in metaverses in the aftermath of the pandemic, researchers have recently conducted a number of studies to understand metaverse usage motivations (Jang, 2022; Oh, 2022; Yu, 2024). In particular, Yu (2024) identified metaverse usage motivations based on an extensive literature review, three rounds of Delphi studies with expert panels, and two online surveys of metaverse users. With this study (Yu, 2024), we sought to explore the moderating effects of four metaverse usage motives, communication, experience of new worlds, self-expression, and economic activity, on the impacts of users' barriers to NFT usage on their metaverse NFT purchase intentions by answering the following research questions.

Research question 1. Is the effect of user barriers (usage, value, risk, tradition, and image) to using metaverse NFTs on metaverse NFT purchase intention moderated by communication as a metaverse usage

Table 2
Prior studies on consumer resistance to technology based on innovation resistance theory.

Researcher(s)	Research topic	Consumer barrier	Dependent variable	Validation result
Lian et al. (2012)	Online shopping	Value →	Intention to use	Adoption (−)
		Tradition →		Adoption (−)
		Value →		Adoption (−)
Lian & Yen (2014)	Online shopping	Value →	Intention to use	Adoption (−)
		Risk →		Adoption (−)
		Tradition →		Adoption (−)
Yu & Chantatub (2015)	Mobile banking	Usage →	Resistance intention	Adoption (+)
		Tradition →		Adoption (+)
		Image →		Adoption (+)
Oktavianus et al. (2017)	Augmented reality mobile games	Usage →	Adoption intention	Adoption (−)
		Value →		Adoption (−)
		Risk →		Adoption (−)
		Tradition →		Adoption (−)
		Image →		Adoption (−)
Gupta & Arora (2017)	Mobile shopping	Usage →	Adoption intention	Adoption (−)
		Risk →		Adoption (−)
		Tradition →		Adoption (−)
		Image →		Adoption (−)
Borraz-Mora et al. (2017)	Internet banking	Value →	Non-adoption intention	Adoption (+)
		Risk →		Adoption (+)
Hew et al. (2019)	Mobile social commerce	Usage →	Intention to use	Adoption (−)
		Value →		Adoption (−)
		Risk →		Adoption (−)
		Tradition →		Adoption (−)
Kaur et al. (2020a)	Mobile payment solutions	Risk →	Recommendation intention	Adoption (−)
		Tradition →	Intention to use	Adoption (−)
		Image →	Recommendation intention	Adoption (−)
	Food delivery applications	Tradition →	Intention to use	Adoption (−)
		Image →	Word of mouth	Adoption (−)
Talwar et al. (2020)	Online travel agencies	Value →	Purchase intention	Adoption (−)
		Risk →		Rejection (+)
Chen et al. (2022)	Mobile ticketing applications	Usage →	Intention to use	Adoption (−)
		Value →		Adoption (−)
		Risk →		Adoption (−)
		Tradition →		Adoption (−)
		Image →		Adoption (−)
Migliore et al. (2022)	Mobile payment	Tradition →	Behavioral intention	Italian: Adoption (−)
Lee & Kim (2022)	Service robots	Usage →	Alternative attractiveness	Adoption (−)
		Tradition →		Adoption (−)
		Image →		Adoption (−)
Chakraborty et al. (2023)	Fitness apps	Usage →	Purchase intention	Adoption (−)
		Risk →		Adoption (−)
		Value →		Adoption (−)
Verma et al. (2023)	Food delivery applications	Image →	Consumer emotion & feeling	Adoption (−)
		Environment→		Adoption (−)
Uddin et al. (2025)	WhatsApp payment system	Usage →	Intention to adopt	Adoption (−)
		Value →		Adoption (−)
		Risk →		Adoption (−)
		Tradition →		Adoption (−)
Nie et al. (2025)	Metaverse products	Usage →	Intention to use	Adoption (−)
		Value →		Adoption (−)
		Tradition →		Adoption (−)

motivation?

Research question 2. Is the effect of user barriers (usage, value, risk, tradition, and image) to using metaverse NFTs on metaverse NFT purchase intention moderated by experience of new worlds as a metaverse usage motivation?

Research question 3. Is the effect of user barriers (usage, value, risk, tradition, and image) to using metaverse NFTs on metaverse NFT purchase intention moderated by self-expression as a metaverse usage motivation?

Research question 4. Is the effect of user barriers (usage, value, risk, tradition, and image) to using metaverse NFTs on metaverse NFT purchase intention moderated by economic activity as a metaverse usage motivation?

3. Method

The research model of this study, developed based on the literature review, is presented in Fig. 2.

3.1. Data collection

To collect the data for the study, we appointed a Korean survey company to administer an online panel survey among 500 respondents aged 10–50 years. Notably, according to the statistics on the age distribution of metaverse users, teenagers account for the highest proportion at 38 % (Pangarkar, 2025). Therefore, prior research on metaverse emphasizes the importance of incorporating the opinions of teenagers when exploring the usage motivations of metaverse users (Yu, 2024), with the largest number of teenage samples being included based on the age distribution of metaverse users (Lee et al., 2023).

The survey company used quota sampling to ensure equal representation of both genders, all age groups, and residential regions. These efforts were made to reduce potential errors in the analysis results that might have arisen due to demographic differences (Yu, 2024). The 500 respondents were evenly split between men and women, with 100

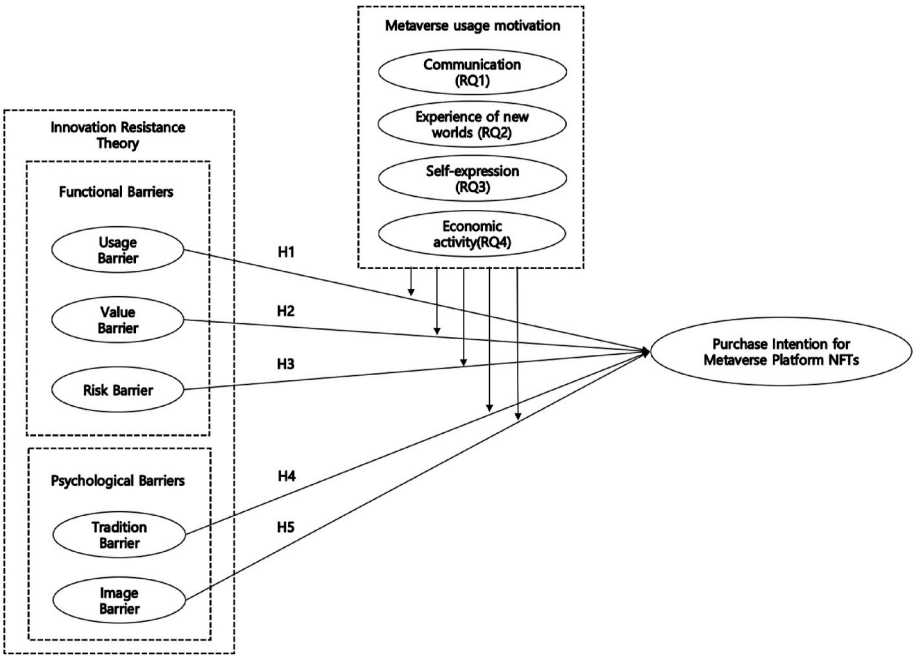


Fig. 2. Research model of this study.

respondents in each age cohort, from teens to 50s. In terms of frequency of metaverse use, 251 respondents (50.2 %) used a metaverse platform 2–3 times a month, 153 (30.6 %) used the platform 4–5 times a month, 57 (11.4 %) used the platform 5–6 times a week or more, and 39 (7.8 %) used the platform several times a day.

We surveyed metaverse platform users who used metaverse at least twice a month but had never purchased NFTs in the metaverse, and we identified these users through a screening question in the study survey. Specifically, respondents who used metaverse less than twice a month or had previously purchased NFTs in the metaverse were automatically disqualified from the online survey. The condition of using metaverse at least twice a month was set to measure the participants' metaverse platform usage motivations, while the condition of never having purchased NFTs was set to measure the participants' barriers to metaverse NFTs. Next, we explained metaverse NFTs to the participants and asked for their perceptions. Additionally, to investigate whether metaverse usage motivations (communication, experience of new worlds, self-expression, and economic activity) moderated the effects of metaverse users' functional and psychological barriers on their metaverse NFT purchase intentions, we measured the participants' metaverse usage motivations. Since our study included teenage respondents, we added an attention check to eliminate dishonest responses from the data and designed the survey to end if the respondent gave the same score consecutively, thereby improving the reliability of the survey.

3.2. Measurement tools

3.2.1. Users' functional and psychological barriers to using metaverse NFTs

To conduct this study, we operationalized user functional and psychological barriers as sources of resistance to the innovation of metaverse NFTs. To measure this construct, the present study adapted items from recent studies that measured consumer resistance barriers based on the framework proposed by Ram and Sheth (1989), modifying them to fit the purpose and context of this research (Appendix 1).

3.2.2. Metaverse usage motivations

We defined metaverse usage motivation as the motivation to fulfill needs in a virtual world where users can interact with each other in economic and cultural activities through advanced immersive technologies using avatars. We measured metaverse usage motivations using a 13-item scale with four factors (communication, experience of new worlds, self-expression, and economic activity) with a factor structure that had been validated earlier (Yu, 2024, Appendix 1).

3.2.3. Metaverse NFT purchase intention

We measured metaverse NFT purchase intention using survey items we adapted from Mackenzie and Lutz (1989) (Appendix 1).

3.3. Analysis methods

We analyzed the study data using SPSS 18.0, AMOS 27.0, and SPSS Process Macro v 4.1 according to the following specific analytical procedures. First, we calculated descriptive statistics, common method bias, reliability analysis, correlation analysis, exploratory factor analysis, and confirmatory factor analysis to examine the reliability and validity of the measurement variables used in this study. Second, we analyzed the relationship between metaverse users' barriers to using metaverse NFTs and their intention to purchase NFTs using structural equation modeling (SEM) to test the research hypotheses. Third, we applied the process macro model 1 to explore the moderating effect of metaverse usage motivation on the relationship between metaverse user barriers to using metaverse NFTs and their metaverse NFT purchase intention. Hayes' (2013) process macro analyzes direct and indirect mediation, moderation, and moderated mediation effects in regression analysis.

4. Results

4.1. Descriptive statistics

We calculated means and standard deviations of the survey responses. We also derived the absolute skewness and kurtosis for the survey items in all survey responses to assess normality. Generally, normality is not exceeded when the absolute skewness, which is the degree of left–right symmetry in the shape of a normal distribution, and kurtosis, which is the shape of the peak of a normal distribution, do not exceed 2 (George & Mallery, 2010). The analysis showed that skewness ranged from −0.619 to 0.833 and kurtosis ranged from −0.616 to 0.509; that is, all values were lower than 2, representing acceptable normality.

4.2. Correlation analysis results

We calculated Pearson's correlation coefficients to measure the correlations between metaverse NFT usage barriers, metaverse usage motivations, and metaverse NFT purchase intention. We found that all correlation coefficients were meaningful at $p < 0.01$. Metaverse NFT usage barriers showed significant negative impacts on the correlation between metaverse usage motivation (−0.12) and metaverse NFT purchase intention (−0.38). The correlation coefficient between metaverse usage motivation and metaverse NFT purchase intention was 0.39, confirming a significant positive relationship (Table 3).

4.3. Confirmatory factor analysis

To measure the construct validity of the measurement model, we conducted confirmatory factor analysis. First, we examined the goodness of fit of the research model; if the model goodness-of-fit indices do not meet the thresholds, the model cannot be adopted irrespective of the hypothesis testing results. All fit indices in this study exceeded the minimum satisfactory standard ($\chi^2 = 574.641$, $df = 172$, $\chi^2/df = 3.341$, GFI = 0.900, IFI = 0.923, TLI = 0.905, CFI = 0.922, and RMSEA = 0.068; Yu & Han, 2021).

To measure the convergent validity, we used three criteria: standardized factor loadings of 0.5 or higher, average variance extracted (AVE) of 0.5 or higher, and composite reliability (CR) of 0.7 or higher (Fornell & Larcker, 1981; Hair et al., 2006; Yu & Han, 2021). The standardized coefficients for metaverse NFT purchase intention and for usage, value, risk, tradition, and image as barriers to metaverse NFT usage were all significant at $p < 0.001$. In addition, AVE exceeded 0.5 and CR exceeded 0.7 for all measured variables except 'value' as a usage barrier. We determined that the study measurement model demonstrated adequate convergent validity (Table 4).

4.4. Structural model

We validated the impacts of users' functional (usage, value, and risk) and psychological (tradition and image) barriers to metaverse NFT use on their metaverse NFT purchase intention and found that usage ($\beta = -0.240$, $p < 0.001$), value ($\beta = -0.140$, $p < 0.05$), risk ($\beta = -0.427$, $p < 0.001$), tradition ($\beta = -0.688$, $p < 0.001$), and image ($\beta = -0.186$, $p < 0.01$) as barriers all had negative effects on metaverse NFT purchase intention (see Table 5 and Fig. 3).

4.5. Moderation analysis

We estimated the moderating effects using a process macro model 1 with the sample size set to 5,000 with bootstrapping and with significance set at 0.05. We calculated the regression equation by dividing the moderating variables into three groups with the mean and ±1 standard deviation, along with the mean centering of the variables, to verify the significance of the simple slope. The results of the analysis are shown in Table 6, but briefly, first, the interaction term between communication as a metaverse usage motivation and usage (coeff. = −0.126, $t = -3.758$, $p < .001$) and value (coeff. = 0.121, $t = 3.276$, $p < .001$) as user barriers to metaverse NFT usage were statistically significant. Second, the interaction terms between experience of new worlds as a metaverse usage motivation and all user barriers to NFTs were not statistically significant. Third, the interaction terms of self-expression as a metaverse usage motivation and usage (coeff. = −0.068, $t = -2.305$, $p < .05$), value (coeff. = 0.100, $t = 3.096$, $p < .01$), risk (coeff. = −0.090, $t = -3.045$, $p < .01$), tradition (coeff. = 0.107, $t = 3.331$, $p < .01$), and image (coeff. = 0.073, $t = 2.105$, $p < .05$) as barriers to using metaverse NFTs were all statistically significant. Finally, the interaction terms of economic activity as a metaverse usage motivation and usage (coeff. = −0.066, $t = -2.523$, $p < .05$) and tradition (coeff. = 0.060, $t = 2.070$, $p < .05$) as barriers to metaverse NFT use were statistically significant. To better understand the moderating role of metaverse usage motivation, see Fig. 4 and Table 7.

Table 3
Correlation analysis results.

	Metaverse NFT barrier	Metaverse usage motivation	Metaverse NFT purchase intention
Metaverse NFT barriers	1		
Metaverse usage motivation	−0.12**	1	
Metaverse NFT purchase intention	−0.38**	0.39**	1

*$p < .05$, **$p < .01$, ***$p < .001$.

5. Discussion and implications

5.1. Discussion of findings

The biggest obstacle to the development and expansion of metaverse NFTs is a shift in user perception. NFTs are literally digital assets that do not exist in the real world and cannot be equated to a real-world resource. In order for users who are accustomed to the existing real economy to accept the innovation of digital assets, accurate analysis of user innovation resistance and marketing communication activities are essential. For the expansion and development of NFTs, it is necessary to change users' perceptions of the value of digital assets and their acceptance of the innovation. Resistance to innovation is not the opposite of innovation adoption but rather the beginning of a process of behavioral change that occurs when consumers are exposed to innovative products. Consumers are often not opposed to innovation but just prefer to postpone their decisions until the innovation has evolved to be common or the product or service has improved (Szmigin & Foxall, 1998). Therefore, identifying the factors that contribute to consumers' resistance to innovation and modifying and improving the products or services to meet consumers' needs and wants can play an important role in the acceptance and diffusion of innovation.

In light of this, we employed the theory of innovation resistance to examine the effects of metaverse users' functional (usage, value, and risk) and psychological (tradition and image) barriers to using metaverse NFTs on their metaverse NFT purchase intention and found that all five barriers had negative effects on NFT purchase intention. The results are discussed in detail below.

First, the negative effect of usage as a barrier to metaverse NFTs on NFT purchase intention was consistent with findings from other research on innovation resistance (Gupta & Arora, 2017; Oktavianus et al., 2017; Sadiq et al., 2021; Sang et al., 2022). Metaverse NFTs have a high barrier to entry based on the understanding of them; most general metaverse users lack knowledge about NFTs and find them complicated. In response, metaverse platform operators need to promote NFTs with clearer information that makes them easier to accurately understand so that users can make informed value determinations.

Second, users' perceptions of the value of metaverse NFTs had a negative effect on their NFT purchase intention, also consistent with other innovation resistance theory studies (Borraz-Mora et al., 2017; Sadiq et al., 2021; Sang et al., 2022; Oktavianus et al., 2017; Talwar et al., 2020). To increase metaverse users' perceptions of the value of metaverse NFTs, it will be necessary for researchers to determine drivers of the perceived value of NFTs in the metaverse so that platform operators can increase metaverse users' perceptions of their value.

Third, users' perceptions of risk related to metaverse NFTs had a negative effect on their NFT purchase intention, which also aligned with other study findings (Borraz-Mora et al., 2017; Gupta & Arora, 2017; Oktavianus et al., 2017; Sadiq et al., 2021; Sang et al., 2022). We believe

Table 4
Confirmatory factor analysis results.

Measurement variable			Measurement item	Unstandardization coefficient	Standardization coefficient	SE	CR	*p*	AVE	CR
Functional barriers	Usage	→	Usage 1	1.000	0.836				0.623	0.831
		→	Usage 2	0.768	0.692	0.049	15.751	0.000***		
		→	Usage 3	0.939	0.832	0.050	18.703	0.000***		
	Value	→	Value 1	1.000	0.526				0.367	0.633
		→	Value 2	1.191	0.617	0.120	9.931	0.000***		
		→	Value 3	1.242	0.667	0.120	10.381	0.000***		
	Risk	→	Risk 1	1.000	0.703				0.516	0.810
		→	Risk 2	1.046	0.683	0.077	13.652	0.000***		
		→	Risk 3	1.197	0.745	0.081	14.743	0.000***		
		→	Risk 4	1.300	0.741	0.089	14.675	0.000***		
Psychological barriers	Tradition	→	Tradition 1	1.000	0.657				0.503	0.801
		→	Tradition 2	1.203	0.734	0.084	14.384	0.000***		
		→	Tradition 3	1.084	0.678	0.081	13.445	0.000***		
		→	Tradition 4	1.287	0.763	0.087	14.857	0.000***		
	Image	→	Image 1	1.000	0.747				0.528	0.817
		→	Image 2	0.972	0.731	0.060	16.107	0.000***		
		→	Image 3	0.948	0.708	0.061	15.576	0.000***		
		→	Image 4	0.975	0.720	0.062	15.840	0.000***		
Purchase intention		→	Purchase intention 1	1.000	0.898				0.664	0.853
		→	Purchase intention 2	1.043	0.892	0.048	21.802	0.000***		
		→	Purchase intention 3	0.754	0.625	0.050	15.091	0.000***		

$^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$. SE: standard error. AVE: average variance extracted. AVE = (∑ Standard loading^2)/[(∑ Standard loading^2)+(∑ Measurement error)]. CR: composite reliability. CR = (∑ Standard loading)^2/[(∑ Standard loading) ^{2} +(∑ Measurement error)].

Table 5
Hypothesis testing results.

Research hypothesis	Channel	Unstandardization coefficient	Standardization coefficient	S.E.	C.R	Verification result
Hypothesis 1	Usage → Purchase intention	−0.243	−0.240	0.047	−5.169***	Supported
Hypothesis 2	Value → Purchase intention	−0.167	−0.140	0.066	−2.522*	Supported
Hypothesis 3	Risk → Purchase intention	−0.438	−0.427	0.059	−7.479***	Supported
Hypothesis 4	Tradition → Purchase intention	−0.721	−0.688	0.070	−10.323***	Supported
Hypothesis 5	Image → Purchase intention	−0.193	−0.186	0.069	−2.798**	Supported

$^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$. SE: standard error. CR: composite reliability.

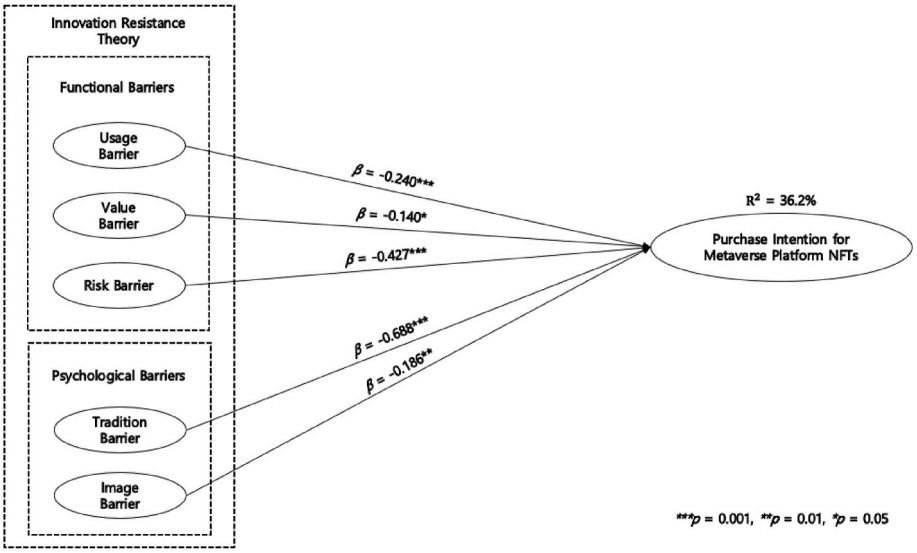


Fig. 3. Results of the structural model testing.

that risk serves as a barrier to metaverse NFT adoption because of recent cryptocurrency crashes. The crash of the cryptocurrencies Lunar and Dogecoin in 2022, followed by the bankruptcy of FTX, the world's fifth-largest margin trading cryptocurrency exchange, have increased users' sense that NFTs are risky. So, legal and institutional safeguards are necessary to protect the possession and value of NFTs so users might perceive less risk associated with their use.

Fourth, users' sense of tradition as a barrier to metaverse NFT use had a negative effect on NFT purchase intention, in keeping with other work (Gupta & Arora, 2017; Oktavianus et al., 2017; Sadiq et al., 2021;

Table 6
Moderation analysis results.

Control variable	Channel	B	*t*	*p*	LLCI	ULCI	Moderation
Communication	Usage → Purchase intention	−0.126	−3.758	0.000***	−0.192	−0.060	Yes
	Value → Purchase intention	0.121	3.276	0.001**	0.048	0.193	Yes
	Risk → Purchase intention	−0.054	−1.609	0.108	−0.120	0.012	No
	Tradition → Purchase intention	0.060	1.753	0.080	−0.007	0.128	No
	Image → Purchase intention	0.040	1.011	0.312	−0.037	0.116	No
Experience of new worlds	Usage → Purchase intention	−0.040	−1.178	0.239	−0.106	0.027	No
	Value → Purchase intention	−0.044	−1.148	0.251	−0.118	0.031	No
	Risk → Purchase intention	0.029	0.903	0.367	−0.034	0.091	No
	Tradition → Purchase intention	−0.032	−0.952	0.342	−0.099	0.035	No
	Image → Purchase intention	−0.030	−0.808	0.419	−0.102	0.043	No
Self-expression	Usage → Purchase intention	−0.068	−2.305	0.022*	−0.127	−0.010	Yes
	Value → Purchase intention	0.100	3.096	0.002**	0.037	0.164	Yes
	Risk → Purchase intention	−0.090	−3.045	0.002**	−0.149	−0.032	Yes
	Tradition → Purchase intention	0.107	3.331	0.001**	0.044	0.171	Yes
	Image → Purchase intention	0.073	2.105	0.036*	0.005	0.142	Yes
Economic activity	Usage → Purchase intention	−0.066	−2.523	0.012*	−0.117	−0.015	Yes
	Value → Purchase intention	0.040	1.305	0.192	−0.020	0.099	No
	Risk → Purchase intention	−0.015	−0.544	0.587	−0.070	0.040	No
	Tradition → Purchase intention	0.060	2.070	0.039*	0.003	0.117	Yes
	Image → Purchase intention	0.046	1.529	0.127	−0.013	0.105	No

*$p < .05$, **$p < .01$, ***$p < .001$. LLCI: lower limit confidence interval. ULCI: upper limit confidence interval.

Sang et al., 2022). The traditional construct of currency has been in operation for centuries, and humanity has established comfort with and trust in traditional currency. It is likely difficult for people to accept such a new currency in a short period of time given that history. To increase trust in NFTs as digital assets with monetary characteristics, it is necessary to provide reliable and well-structured information about NFTs through educational initiatives and public awareness campaigns, which can help foster the perception that NFTs possess economic value and utility.

Fifth, users' image of NFTs as a barrier to their use had a negative effect on their purchase intention of NFTs, which was also consistent with earlier innovation resistance research (Gupta & Arora, 2017; Oktavianus et al., 2017; Sadiq et al., 2021). Communication with metaverse users can improve their perceptions of NFTs as a new technology, for instance regarding policies on refunds and exchanges, which have proven to be a major issue for users (Yu & Han, 2021). Metaverse platform operators can improve the image of NFTs by making it clear to consumers that they will be responsive to any and all concerns, including friendly customer service and smooth transactions and problem handling.

Next, we explored whether the impacts of users' functional and psychological barriers to using metaverse NFTs on their metaverse NFT purchase intentions would be moderated by their metaverse usage motivations, specifically, communication, experience of new worlds, self-expression, and economic activity. The main findings are discussed below. First, the negative impact of usage as an NFT usage barrier on NFT purchase intentions increased as communication, self-expression, and economic activity as metaverse use motivations increased, and the negative impact of risk increased as the self-expression motivation increased. This means that the greater the motivation to use the metaverse for communication, self-expression, and profit, the greater the impacts of the users' barriers to using NFTs on reducing their NFT purchase intentions. These results suggest that it is important for metaverse platform operators to accurately identify their users' motivations for using their platforms in order better tailor the user experience to overcome perceived barriers to both NFT market entry and NFT purchase intention.

Second, the negative impact of value as a barrier to metaverse NFT use on NFT purchase intention was moderated by higher communication and self-expression as motivations to use a metaverse; negative perceptions of the value of NFTs in the metaverse had weaker impacts on NFT purchase intention when participants had greater motivation to use the metaverse for communication and self-expression. In addition, the negative impact on NFT purchase intention of tradition as a barrier was moderated as the self-expression and economic activity motives increased, and the negative effect on NFT purchase intention of image (for instance the image of metaverse operators as untrustworthy customer service agents in the event of a problem) as a barrier to NFT use decreased as self-expression increased as a motivation.

The sum of these results indicates that if metaverse platform operators find that their users are resistant to using NFTs on the platform, operators can tailor effective messaging to their users by understanding their motivations for using the platforms; users who come for self-expression will probably be differently motivated from users who

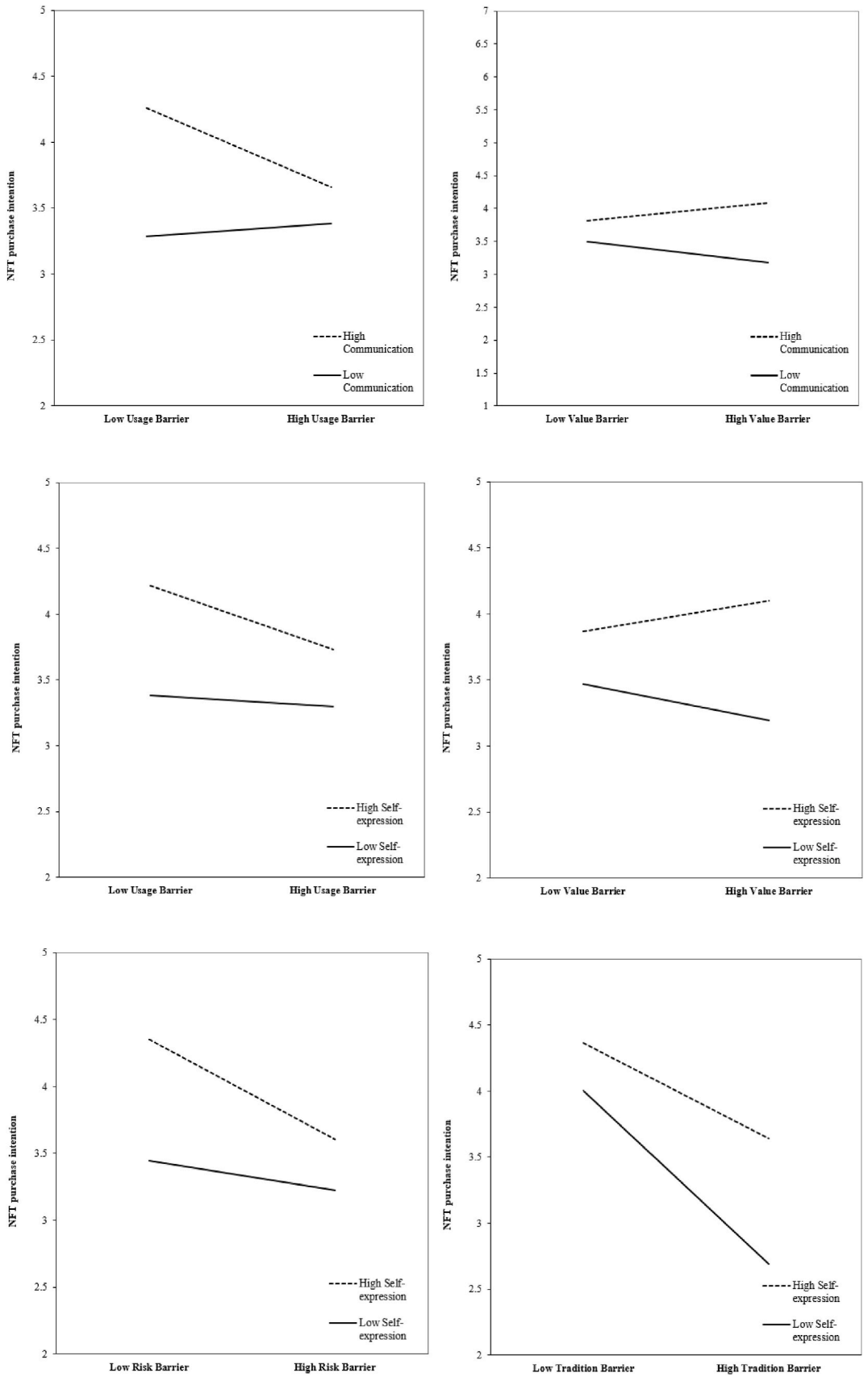


Fig. 4. Usage motivation moderation effects.

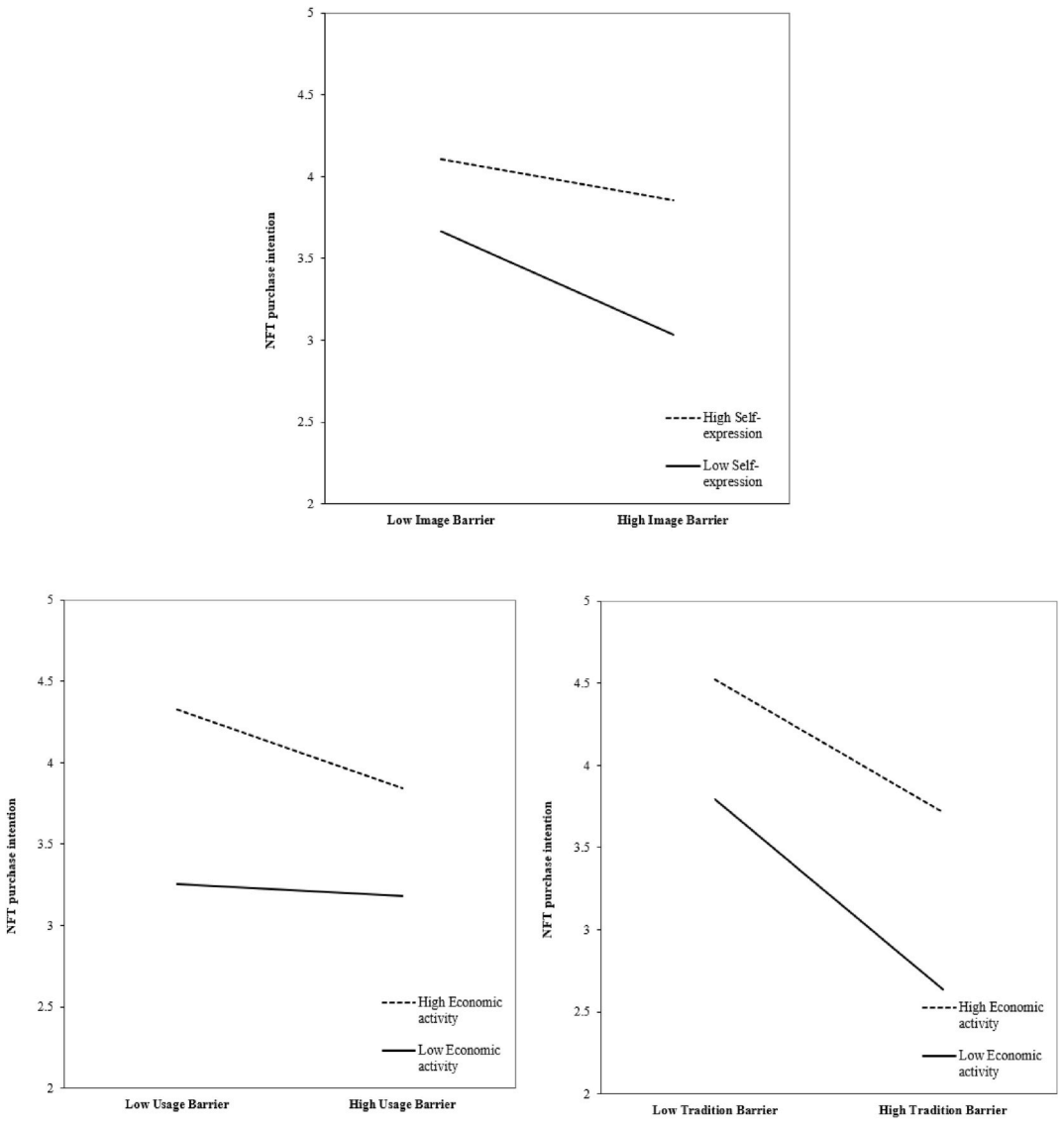


Fig. 4. (*continued*).

Table 7
Metaverse usage motivation moderation effects by user barrier.

Independent variable	Channel	Results	Reduced purchase intention
Usage	Usage → NFT purchase intention	O	Reinforce as communication motivation grows
	Usage → NFT purchase intention	O	Reinforce as self-expression motivation grows
	Usage → NFT purchase intention	O	Reinforce as economic activity motivation grows
Value	Value → NFT purchase intention	O	Relieve as communication motivation grows
	Value → NFT purchase intention	O	Relieve as self-expression motivation grows
Risk	Risk → NFT purchase intention	O	Reinforce as self-expression motivation grows
Tradition	Tradition → NFT purchase intention	O	Relieve as self-expression motivation grows
	Tradition → NFT purchase intention	O	Relieve as economic activity motivation grows
Image	Image → NFT purchase intention	O	Relieve as self-expression motivation grows

come to engage in economic activities on the platform. Additionally, different user barriers to using NFTs have differing impacts depending on those usage motivations, so again, understanding what perceptions cause users to hesitate to use NFTs on a metaverse platform can help platform operators tailor their messaging to expand NFT use. For example, if your users have a highly negative perception of the value of NFTs, you should develop content that allows users to earn NFTs and use them to their advantage throughout their metaverse experiences, thereby increasing their perceived value. Or if users perceive metaverse NFTs as risky, operators can offer clear messaging that issues with NFTs will be promptly resolved; they could also ensure that users know they will not face unnecessary exposure from incorporating NFTs into their self-expression and economic activities with their avatars on the platform.

Finally, the following findings from our exploration of the moderation effects of metaverse motivation support the earlier findings of some researchers in innovation resistance theory that resistance and adoption can simultaneously exist (Hew et al., 2019; Laukkanen & Kivinemi, 2010; Ram, 1987; Talwar et al., 2020). First, we observed that value as a barrier had a positive effect on NFT purchase intention in the group with high metaverse communication and self-expression motivation. Second, we found a moderation effect of metaverse user motivations

(communication, self-expression, and economic activity) between the effects of user barriers (value, tradition, and image) on the decline in NFT purchase intention. In the literature (Hew et al., 2019; Talwar et al., 2020), the phenomenon of user barriers being positively associated with purchase intentions has been described as the privacy paradox. The privacy paradox refers to the coexistence of privacy concerns and usage intentions (Taddicken, 2014). In a research study on consumer innovation resistance to mobile social commerce among social media users, Hew et al. (2019) inferred that users with high privacy concerns might perceive mobile social commerce as a useful service and thus be able to justify the potential loss of privacy. They found that social media users with high privacy concerns do not apply their concerns to their usage intention, and they argued that social media users intend to use social media despite having high privacy concerns. We designed this study for twice-monthly metaverse users who have never purchased metaverse NFTs. Therefore, we infer from our analysis that metaverse users can recognize the metaverse as a useful service. We also propose a theoretical framework for understanding the motivations for using the metaverse as a way to strengthen the possibility of coexisting resistance to and adoption of metaverse NFTs.

5.2. Theoretical implications

We propose the following implications based on our findings. First, this study has academic significance in that we provided a theoretical framework for examining the relationships between users' functional and psychological barriers to using metaverse NFTs, their metaverse NFT purchase intentions, and their motivations for using metaverse platforms. Since NFTs could form the basis of the future metaverse economic system, this is important information. Researchers on innovation resistance theory have already identified functional and psychological barriers to innovation adoption in various fields, such as online shopping (Lian et al., 2012; Lian & Yen, 2014), mobile banking (Yu & Chantatub, 2015), augmented reality mobile games (Oktavianus et al., 2017), mobile shopping (Gupta & Arora, 2017), internet banking (Borraz-Mora et al., 2017), mobile social commerce (Hew et al., 2019), online travel agencies (Talwar et al., 2020), mobile payment solutions (Kaur et al., 2020a), food delivery applications (Kaur et al., 2020b; Verma et al., 2023), mobile ticketing applications (Chen et al., 2022), mobile payment (Migliore et al. (2022), service robots (Lee et al., 2022), fitness apps (Chakraborty et al. (2023), WhatsApp payment system (Uddin et al., 2025), and metaverse products (Nie et al., 2025). However, we extended the scope and application of innovation resistance theory by examining functional and psychological barriers to currency innovation in the context of digital assets, NFTs.

Second, the study is valuable in that we examined the moderation effect of motivation to use a metaverse on the effects of functional and psychological barriers to metaverse NFT use on users' metaverse NFT purchase intention. In addition, our exploration led to one conclusion: if metaverse operators are encountering resistance from their users to purchase and use NFTs in the virtual world, understanding why users use their platforms and what aspects of NFT adoption they find risky enough to pose barriers to usage will allow them to tailor messaging and promotions to different populations of platform users. A user with high self-expression motivation and privacy concerns needs to know that using NFTs in the metaverse will not put them at risk for any privacy violations such as identify theft. On the other hand, a user who is motivated by the economic activity options in the metaverse might resist NFT adoption because of the perception that they lack value; this user will need different messaging.

5.3. Practical implications

This study provides the following practical implication: combining metaverse platforms and NFTs can make a great contribution to expanding the metaverse platform market. The macro profit model of metaverse platforms is B2B, which is to build and sell metaverse platforms to enterprises, rent space, and advertise. However, for a company to enter the metaverse platform or commission their own advertisements, a large number of users need to access the platform. To attract a large number of users, it is necessary to accurately identify and analyze user motivations to establish the direction of platform development. NFT publication and sales can also have a significant role in the micro profit model of the metaverse platform in the form of B2C and C2C; users can generate revenue on their own by issuing and selling NFTs on the metaverse platform or through commissions from production and transaction activities between users.

If consumers perceive that adopting an innovation will bring great change, they will resist the innovation, and to address this resistance, companies will need to modify the innovation to meet consumer needs. Through this feedback loop, the innovation is ultimately adopted or rejected (Ram, 1987). This means that metaverse platform providers need to accurately understand users' resistance to innovation and modify their innovations to match consumer desires. By applying Ram's (1987) innovation resistance model to metaverse users' barriers to using metaverse NFTs, which we demonstrated negatively affected users' NFT purchase intentions, we present a virtuous cycle of metaverse platform user innovation adoption in Fig. 5.

The cycle in Fig. 5 operated as follows. First, identify user needs by understanding their motivations for using a particular metaverse platform. Second, determine the psychological and functional barriers that keep users from engaging with platform NFTs. Third, revise the metaverse platform to address the users' concerns, meet their needs, and reduce their perceptions of barriers to NFT adoption. Fourth, measure user response to platform updates through various indicators and surveys. Fifth, repeat the process if the innovation is still not adopted. Notably, in the case of this study, even if users adopt the innovation, the NFTs, it is still necessary for platform operators to maintain communication with users to ensure that their needs are always being met.

5.4. Limitations and future research avenue

Despite the meaningful findings and contributions of this study, several limitations warrant careful consideration.

First, the data were collected from a sample of 500 users residing in South Korea. Although demographic balance was ensured through quota sampling, cultural and societal factors may exert significant influence on both innovation resistance and metaverse usage motivations. Accordingly, the generalizability of the findings to broader global populations is inherently constrained. Future studies should consider cross-cultural comparative designs to explore whether user resistance varies across different socio-cultural and institutional contexts.

Second, this study conceptualized metaverse users as a homogeneous group, without distinguishing among specific platform environments. However, various metaverse platforms such as Roblox, Zepeto, Fortnite, and Meta Horizon offer distinct technological affordances, levels of immersion, and NFT-related functionalities. These platform-specific characteristics may lead to divergent patterns of innovation resistance and motivational drivers. Subsequent research should adopt a more granular, platform-level analytical approach to derive tailored insights that can inform targeted interventions by platform developers and marketers.

Third, NFTs were treated as a unitary construct, despite the heterogeneity of NFT types and use cases. NFTs can represent a wide range of digital assets, including avatar items, virtual real estate, digital art, and event-based access tokens. Each NFT category may vary in terms of perceived value, usability, and associated risk, which could differentially influence users' resistance and purchase intentions. Future research should disaggregate NFT types to examine their unique effects on user perception and adoption behaviors.

Beyond these methodological considerations, the future trajectory of the metaverse and NFTs demands broader theoretical reflection. For all

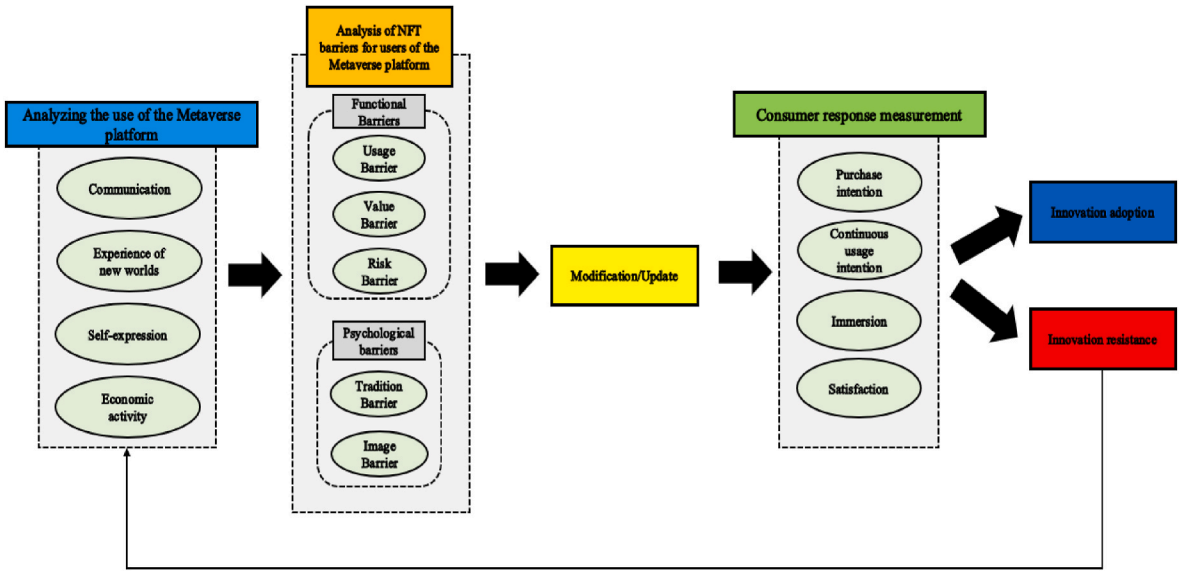


Fig. 5. Virtuous cycle of innovation adoption by metaverse platform users.

time, humanity has been constantly exploring new spaces and dreaming of migrating to new placeslands, such as space, the ocean floor, and beyond. Through the Internet, our living space has moved from offline to online, and just as the online space has moved from the Internet to online games, smartphones, and social media, the day will come when the human living space will move to the ideal metaverse beyond our dreams as technology advances. Although the growth of the metaverse and metaverse NFTs appears to have temporarily slowed at present, metaverse technology developments remain an inevitable reality. Therefore, it is essential to build legal and institutional frameworks grounded in continuous research, development, and public discourse in order to realize the ideal metaverse that users envision.

In this context, the Gartner Hype Cycle serves as an insightful conceptual framework for understanding the lifecycle of emerging technologies such as the metaverse and NFTs. The Gartner Hype Cycle, introduced by Gartner Inc., outlines the typical trajectory of emerging technologies by illustrating how public expectations and perceived value evolve over time (Dedehayir & Steinert, 2016). According to the model, technologies typically undergo a series of stages, beginning with an initial surge of optimism and exaggerated expectations, followed by a period of disillusionment, and eventually culminating in a gradual restoration of realistic expectations (Fenn & Raskino, 2008). The recent rise and subsequent decline in public and market interest in the metaverse and NFTs can be interpreted as part of this typical innovation cycle, progressing from a phase of inflated expectations through a trough of disillusionment, and advancing toward a slope of enlightenment and a plateau of productivity. This interpretation supports the theoretical view that innovation resistance does not necessarily indicate rejection but may instead represent a transitional stage in the broader diffusion process.

In fact, Gartner forecasts that by 2026, approximately one-quarter of the global population will engage with the metaverse for at least an hour daily for purposes such as work, education, or leisure, and that 30 % of global enterprises will be offering products and services compatible with metaverse environments (Chan et al., 2024). Reflecting this projected expansion, academic interest has also intensified; during the peak of the industry's hype cycle, scholarly attention to the metaverse surged, with the number of related publications indexed by Google Scholar increasing fourfold—from approximately 500 to 2,000 (Ng, 2022; Park et al., 2025). Together, these trends underscore the growing relevance of understanding user attitudes and resistance in the early stages of metaverse and NFT diffusion.

To build on this perspective, future research could benefit from integrating the Gartner Hype Cycle with Innovation Resistance Theory and metaverse usage motivations to empirically identify the perceived stage of NFTs within the hype cycle, as recognized by metaverse users and stakeholders. Investigating how such perceptions influence adoption behavior and resistance would offer meaningful theoretical and practical insights. Such an approach could facilitate a more nuanced diagnosis of key barriers in the innovation adoption process of metaverse NFTs, ultimately supporting the sustainable diffusion of digital innovations within the emerging virtual economy.

6. Conclusion

This study applied innovation resistance theory to explore the impact of metaverse users' innovation resistance factors toward metaverse NFTs on their purchase intention in the context of ongoing debates regarding the future of the metaverse and metaverse NFTs. Additionally, this study examined the moderating effect of metaverse usage motivation on the relationship between innovation resistance factors and purchase intention. By doing so, it fills a gap in existing literature, which has predominantly focused on the technological aspects of the metaverse and metaverse NFTs, and expands the scope of innovation resistance theory.

The findings confirm that functional barriers (usage, value, and risk barriers) and psychological barriers (tradition and image barriers) negatively impact metaverse NFT purchase intention. Furthermore, among the metaverse usage motivations, communication, self-expression, and economic activity were found to moderate the effects of these barriers.

This study contributes to the understanding of how metaverse users perceive the innovation that is metaverse NFTs. It also presents concrete strategies for metaverse platforms to manage resistance barriers when issuing NFTs based on users' metaverse usage motivations. Moreover, it suggests countermeasures to strengthen users' metaverse usage motivations to reduce the impact of resistance barriers when metaverse-issued NFTs face innovation resistance. Based on these discussions, this study proposes a virtuous cycle of innovation adoption by metaverse platform users, grounded in Ram's innovation resistance model.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix 1. Measurement items used in the study

Variable (Cronbach's a)	Measurement item	Reference
Usage (0.829)	NFTs are inconvenient because they're not always available.	, ; Lian et al. (2012); Sadiq et al. (2021); van Klyton et al. (2021)
	NFTs are inconvenient because they can't be used everywhere.	
	NFTs are inconvenient because they're complicated to understand.	
Value (0.758)	NFTs are uneconomic compared to traditional currencies.	
	NFTs have low value for money.	
	NFTs are priced and traded by speculation, regardless of value.	
Risk (0.808)	Afraid that possession of NFTs will not be authentically recognized	
	Afraid that the price of an NFT will plummet after I buy it	
	Afraid that the metaverse platform will shut down after I buy NFTs	
	Concerns about the lack of legal and institutional protections for NFTs.	
Tradition (0.800)	I don't trust NFTs.	
	I don't need NFTs; existing currencies are good enough.	
	NFTs are unstable compared to traditional currencies.	
	It's hard to get trustworthy information about NFTs.	
Image (0.816)	If a problem occurs, it is unlikely that the NFT service provider will be able to resolve it smoothly.	
	It's difficult to communicate with the NFT provider or customer support if a problem occurs	
	I do not believe that NFTs currently being sold in the metaverse are real products or currencies.	
	I have doubts about NFTs.	
Communication (0.826)	Using the metaverse to communicate and socialize with others	Yu (2024)
	Using the metaverse to socialize with friends	
	Using the metaverse to connect with people without contact	
	Using the metaverse because it allows me to transcend space and meet and connect with people around the world	
Experience of new worlds (0.808)	Using the metaverse because it allows me to experience a new life without the constraints of time and space, unlike my real life	
	Using the metaverse because it allows me to indirectly experience places I can't physically go	
	Using the metaverse because it allows me to create worlds that wouldn't be possible in reality	
Self-expression (0.807)	Using the metaverse because it allows me to represent the ideal version of myself through an avatar	
	Using the metaverse because it allows me to customize my avatar with different items	
	Using the metaverse because it allows me to express my multiple identities through my avatar	
Economic activity (0.783)	Using the metaverse because it allows me to make money from user-generated content and products	
	Using the metaverse because it allows for new economic activities through jobs that are different from my real life	
	Using the metaverse because it allows users to conduct transactions	
Metaverse NFT purchase intention (0.838)	I will buy NFTs in the metaverse.	Mackenzie & Lutz (1989);
	I would buy NFTs in the metaverse.	
	I might buy NFTs in the metaverse.	

Data availability

Data will be made available on request.

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